

## ⇒ LITERALS &amp; LEXICAL

`#t` `#f` booleans (also `#true` `#false`); only `#f` is false  
`#\a` `#\space` `#\x41` characters — `#\x41` is `#\A`  
`"a\n b"` string; escapes `\n` `\t` `\\"` `\\` `\x41`;  
`#"C:\path"` raw string — no escapes (SRFI 267)  
`42` `-7` `1/3` `6.02e23` exact ints and rationals, floats  
`#x1F` `#b1010` `#o17` hex, binary, octal  
`#e1.5` `#i1/3` force exact / inexact  
`'foo` `|two words|` symbols; `|...|` allows any chars  
`#{1 2 3}` vector literal — immutable  
`#u8(1 2 255)` bytevector literal  
`; line` `#| block` `|#` `#;`(datum) comment forms  
Names: `pred?` asks, `mut!` mutates, `a->b` converts. On this sheet `;>` means “evaluates to”.

## ⇒ DEFINE &amp; LAMBDA

`(define x expr)` bind a value  
`(define (f a b) body ...)` define a procedure  
`(define (f a . rest) body ...)` rest args as a list  
`(lambda (a b) body ...)` anonymous procedure;  
`(lambda args ...)` takes all args as a list  
`(define-values (a b) expr)` bind multiple values  
`(case-lambda (formals body ...) ...)`  
 dispatch on arity [case-lambda]  
`(set! var expr)` assign an existing binding

```
(define (add a b) (+ a b))
(add 2 3) ;=> 5
```

## ⇒ BINDING FORMS

`(let ((x 1) (y 2)) body ...)` parallel bindings  
`(let* ((x 1) (y (+ x 1))) ...)` sequential — each sees the previous  
`(letrec ((f ...) (g ...)) ...)` mutually recursive (also `letrec*`)  
`(let-values (((a b) expr) ...)) ...)` bind multiple values (also `let*-values`)

```
(let loop ((n 5) (acc 1)) ; named let
  (if (= n 0) acc
      (loop (- n 1) (* n acc)))) ;=> 120
```

Tail calls run in constant stack — recurse freely; a named `let` is the idiomatic loop.

## ⇒ CONDITIONALS &amp; FLOW

`(if test then else)` two-way branch; `else` optional  
`(cond (test expr ...) ... (else ...))` first true clause wins; `(test => proc)` passes the value on  
`(case key ((d1 d2) e ...) (else ...))` dispatch on `eqv?` against datum lists  
`(and e ...)` (or `e ...`) short-circuit; return the deciding value  
`(when test body ...)` one-armed if (and unless)  
`(begin e ...)` sequence; value of the last  
`(do ((v init step) ...) (test res) body ...)` general loop

```
(case (* 2 3)
  ((2 3 5 7) 'prime)
  (else 'composite)) ;=> composite

(do ((i 0 (+ i 1))
      (sum 0 (+ sum i)))
    ((= i 5) sum)) ;=> 10
```

## ⇒ QUOTE &amp; QUASIQUOTE

`'(a b c)` quoted literal data  
``(1 ,x ,@xs)` template: `,` inserts a value, `,@` splices a list  
`(let ((x 2) (xs '(3 4))) `(1 ,x ,@xs))`  
`;> (1 2 3 4)`

## ⇒ MACROS

`(define-syntax name (syntax-rules (lits) (pattern template) ...))` hygienic rewrite rules; ... repeats the preceding pattern  
`(let-syntax ...) (letrec-syntax ...)` local macros  
`(syntax-error msg args ...)` fail at expansion time

```
(define-syntax my-when
  (syntax-rules ()
    ((my-when test body ...)
     (if test (begin body ...)))))
```

## ⇒ RECORDS

```
(define-record-type <point>
  (make-point x y) ; constructor
  point? ; predicate
  (x point-x) ; getter
  (y point-y set-point-y!))

(point-x (make-point 3 4)) ;=> 3
```

## ⇒ EXCEPTIONS

`(error msg irritant ...)` raise an error object  
`(guard (e (test expr ...) ... (else ...)) body ...)` catch: first clause matching `e` handles it  
`(raise obj)` `(raise-continuable obj)` raise any value; continuable handlers may return a replacement  
`(with-exception-handler handler thunk)` install a handler around `thunk`  
`(error-object-message e)`  
`(error-object-irritants e)` inspect (also `error-object?`)

```
(guard (e (#t (error-object-message e)))
  (error "boom")) ;=> "boom"
```

## ⇒ MULTIPLE VALUES

`(values a b ...)` return several values  
`(call-with-values producer consumer)` consume them  
`(let-values (((a b) (values 1 2))) (+ a b)) ;=> 3`

## ⇒ CONTINUATIONS

`(call/cc proc)` capture the continuation — escape or resume  
`(call/ec proc)` one-shot escape, much cheaper (Kaappi)  
`(dynamic-wind before thunk after)` guaranteed entry/exit around `thunk`

## ⇒ PARAMETERS &amp; PROMISES

`(make-parameter init)` dynamic variable, called to read  
`(parameterize ((p v) ...) body ...)` rebind within body  
`(delay expr)` promise; `(force p)` runs once, memoizes [lazy]

## ⇒ LIBRARIES

```
(define-library (my lib)
  (import (scheme base))
  (export hello)
  (begin
    (define (hello) 'hi)))

(import (only (srfi 1) fold)) selective
import
(import (prefix (srfi 1) s:)) prefixed; also
(rename ... (old new)) and (except ...)
(cond-expand (kaappi ...) (else ...))
feature-conditional code
(include "file.scm") splice a source file
(my lib) loads from my/lib.sld on the library path.
```

## ⇒ KAAPPI CLI

`kaappi repl` · `kaappi f.scm` runs · reads stdin  
`kaappi compile f.scm -o app` native binary via LLVM  
`kaappi check f.scm` static analysis; KP-code lints  
`kaappi test tests/` SRFI-64 runner, one subprocess per file  
`kaappi fmt --check` canonical formatter (2-space R7RS)  
`kaappi explain KP4001` what a diagnostic means, with fix  
`kaappi doctor · features --json` install self-check · build info  
`--lib-path DIR --sandbox --timeout MS` key flags; sandbox blocks FFI, files, eval  
 Install: `curl -fsSL https://kaappi-lang.org/install.sh | bash`

## ⇒ REPL

`kaappi>` prompt; unbalanced parens continue on the next line  
`_` the last result  
`,help` `,quit` `,version` meta-commands start with a comma  
`,time e` `,type e` `,describe s` `,apropos str` inspect  
`,expand e` `,dis e` `,profile e` `macro-expand` · `bytecode` · `profile`  
`,import (srfi 1)` `,load f.scm` bring in libraries and files  
`,break f` `,step e` debugger: `step` `next` `continue` `locals` `backtrace`  
`~/kaappi/history` · `~/kaappi/config` persistence; Ctrl-D exits

## ⇒ THOTTAM · PACKAGES

`thottam install kaappi-web` with deps, into `~/kaappi/lib`  
`thottam install PKG@v1.0.0` pin a tag; ranges: `@">=0.2.0"`  
`thottam install PKG:URL` install from any git URL  
`thottam remove · list · update [PKG]` manage installed packages  
`thottam verify · --locked` install check/enforce `~/kaappi/thottam.lock`  
`kaappi.pkg` manifest: name version depends build  
 Libraries are plain `.sld` files — import after install: `(import (kaappi web))`.

## ⇒ NUMBERS

(+ a ...) (- a b ...) (\* a ...) (/ a b ...) arithmetic; exact / yields rationals  
 (/ 1 3) **⇒ 1/3**  
 (expt 2 64) **⇒ 18446744073709551616**  
 (= a b ...) (< ...) (> ...) (<= ...) (>= ...) numeric compare, chainable  
 zero? positive? negative? odd? even? sign and parity  
 (abs n) (min a ...) (max a ...) (quotient a b) (remainder a b) (modulo a b) integer division; modulo follows the divisor's sign  
 (floor/ a b) (truncate/ a b) two-value division (also floor-quotient etc.)  
 (gcd a ...) (lcm a ...) (floor x) (ceiling x) (truncate x) (round x) round halves to even  
 (expt a b) (square a) (exact-integer-sqrt n) powers; integer sqrt with remainder  
 (sqrt x) (exp x) (log x b?) (sin x) (cos x) (tan x) (atan y x?) [inexact]  
 finite? infinite? nan? float predicates [inexact]  
 (exact x) (inexact x) convert exactness  
 (exact 0.5) **⇒ 1/2**  
 (numerator q) (denominator q) (rationalize x tol) rational parts  
 (number->string n radix?) (string->number s radix?)  
 (number->string 255 16) **⇒ "ff"**  
 (string->number "ff" 16) **⇒ 255**  
 number? integer? rational? real? complex? tower predicates  
 exact? inexact? exact-integer? exactness predicates

## ⇒ EQUIVALENCE

(eq? a b) identity — symbols, small ints, same object  
 (eqv? a b) eq? plus numbers and chars by value  
 (equal? a b) deep structural equality  
 (equal? '(1 2) (list 1 2)) **⇒ #t**  
 (eqv? 1.0 1) **⇒ #f**  
 (not x) #t only for #f  
 (boolean=? a b ...) (symbol=? a b ...)

## ⇒ TYPE PREDICATES

number? string? char? symbol? boolean? atoms  
 pair? null? list? vector? bytevector? compound  
 procedure? port? eof-object? promise? other

## ⇒ PAIRS &amp; LISTS

(cons a d) (car p) (cdr p) build·first·rest  
 (cons 1 '(2 3)) **⇒ (1 2 3)**  
 (set-car! p v) (set-cdr! p v) mutate a pair  
 (list a ...) (make-list n fill?)  
 (length ls) (reverse ls) (append ls ...) (list-ref ls k) (list-tail ls k)  
 (list-set! ls k v) zero-based access  
 (map f ls ...) (for-each f ls ...) apply elementwise / for effect  
 (map (lambda (x) (\* x x)) '(1 2 3)) **⇒ (1 4 9)**  
 (member x ls) (memv x ls) (memq x ls) find tail by equal?/eqv?/eq?  
 (assoc k al) (assv k al) (assq k al) association-list lookup  
 (assq 'b '((a 1) (b 2))) **⇒ (b 2)**  
 (list-copy ls) shallow copy  
 (caar p) (cadr p) ... (cddddr p) compound accessors [cxr]

## ⇒ STRINGS

(string ch ...) (make-string n ch?) (string-length s) (string-ref s k) by codepoint — UTF-8 native  
 (string-set! s k ch) strings are mutable  
 (substring s a b) half-open [a,b)  
 (substring "hello" 1 4) **⇒ "ell"**  
 (string-append s ...) (string->list s) (list->string ls) also string->vector, vector->string  
 (string-copy s a? b?) (string-copy! to at from) (string-fill! s ch)  
 (string-map f s ...) (string-for-each f s ...) string=? string<? string>? string<=? string>=? lexicographic compare  
 string-ci=? ... string-upcase string-downcase string-foldcase case ops [char]  
 (string->symbol s) (symbol->string sym)

## ⇒ CHARACTERS

(char->integer c) (integer->char n) codepoint conversion  
 (char->integer #\A) **⇒ 65**  
 char=? char<? char>? char<=? char>=? compare (char-ci... [char])  
 char-alphabetic? char-numeric? char-whitespace? classify [char]  
 char-upper-case? char-lower-case? [char]  
 (char-upcase c) (char-downcase c) (char-foldcase c) [char]  
 (digit-value c) digit as number, else #f [char]

## ⇒ VECTORS

(vector a ...) (make-vector n fill?) (vector-length v) (vector-ref v k) (vector-set! v k x)  
 (vector->list v a? b?) (list->vector ls)  
 (vector->list #(1 2 3) 1) **⇒ (2 3)**  
 (vector-copy v a? b?) (vector-copy! to at from) (vector-fill! v x)  
 (vector-append v ...) (vector-map f v ...) (vector-for-each f v ...) (vector-map + #(1 2) #(10 20)) **⇒ #(11 22)**

## ⇒ BYTEVECTORS

(bytevector b ...) (make-bytevector n fill?) bytes 0–255  
 (bytevector-length bv) (bytevector-u8-ref bv k) (bytevector-u8-set! bv k b)  
 (bytevector-copy bv a? b?) (bytevector-copy! to at from) (bytevector-append bv ...) (utf8->string bv a? b?) (string->utf8 s a? b?) (utf8->string #u8(104 105)) **⇒ "hi"**

## ⇒ CONTROL

(apply f a ... args) call with a list as the final args  
 (apply + 1 '(2 3)) **⇒ 6**  
 (read-error? e) (file-error? e) classify raised errors

## ⇒ PORTS &amp; I/O

(display x port?) (write x port?) human-readable vs re-readable [write]  
 (newline port?) (write-char c) (write-string s)  
 (read port?) read one datum [read]  
 (read-line port?) (read-char port?) (read-string k port?) (read-line (open-input-string "one")) **⇒ "one"**

(peek-char port?) look ahead without consuming  
 (read-u8) (write-u8 b) (read-bytevector k) (write-bytevector bv) binary I/O  
 (eof-object) (eof-object? x) end-of-input value  
 (open-input-string s) (open-output-string) string ports;  
 (get-output-string p) extracts (read (open-input-string "(1 2)")) **⇒ (1 2)**  
 (call-with-input-file f proc) (call-with-output-file f proc) auto-close [file]  
 (with-input-from-file f thunk) (with-output-to-file f thunk) redirect current ports [file]  
 (open-input-file f) (open-output-file f) [file]  
 (close-port p) (call-with-port p proc) (flush-output-port p?) (current-input-port) (current-output-port) (current-error-port) default ports, parameterizable  
 (file-exists? f) (delete-file f) [file]  
 port? input-port? output-port? textual-port? binary-port? predicates

## ⇒ SYSTEM &amp; EVAL

(exit code?) terminate [process-context]  
 (command-line) program args as a list [process-context]  
 (get-environment-variable name) one env var; -variables for all [process-context]  
 (current-second) (current-jiffy) (jiffies-per-second) wall clock·monotonic ticks [time]  
 (features) feature symbols — pairs with cond-expand  
 (load "f.scm") [load]  
 (eval expr env) (environment import-set ...) [eval]  
 (real-part z) (imag-part z) (magnitude z) (angle z) complex accessors [complex]

## ⇒ HASH TABLES · SRFI 69

(import (srfi 69)) — for insertion-ordered iteration use SRFI 250 instead.  
 (make-hash-table equal? hash?) create; equality and hash optional  
 (hash-table-set! ht k v) associate key with value  
 (hash-table-ref ht k fail?) look up; calls fail thunk on a miss  
 (hash-table-ref/default ht k default) look up with fallback  
 (hash-table-update!/default ht k f default) update via f  
 (hash-table-delete! ht k) (hash-table-exists? ht k) (hash-table-size ht) (hash-table-keys ht) (hash-table-values ht)  
 (hash-table->alist ht) (alist->hash-table al) convert; also hash-table-walk, hash-table-copy, hash-table?

## ⇒ BEYOND R7RS-SMALL

(import (srfi 1)) list library: fold filter reduce iota any every delete-duplicates ...  
 (import (srfi 18)) OS threads; green fibers are built in  
 600+ built-in procedures, 170+ SRFIs, JSON · HTTP · SQLite · Postgres · Redis via thottam packages ⇒ [kaappi-lang.org/procedures/](http://kaappi-lang.org/procedures/)